



Doctor of Philosophy in Computer Applied Technology

Semiannual Progress Report

2025-07

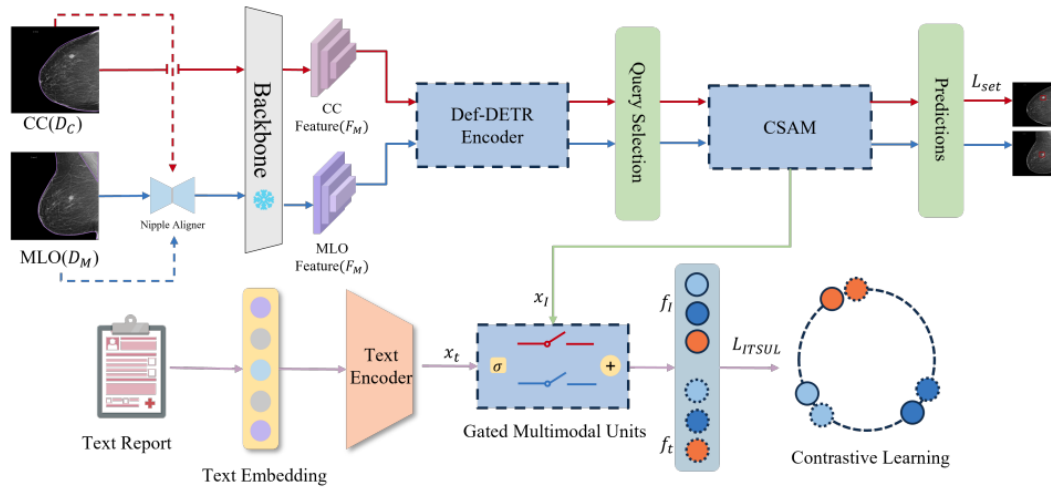
Student

Name	Yixuan Zhang	ID	P2425464
Supervisor	Dr. Tao Tan		
Admission Cohort	2024-09	Months of Study	6

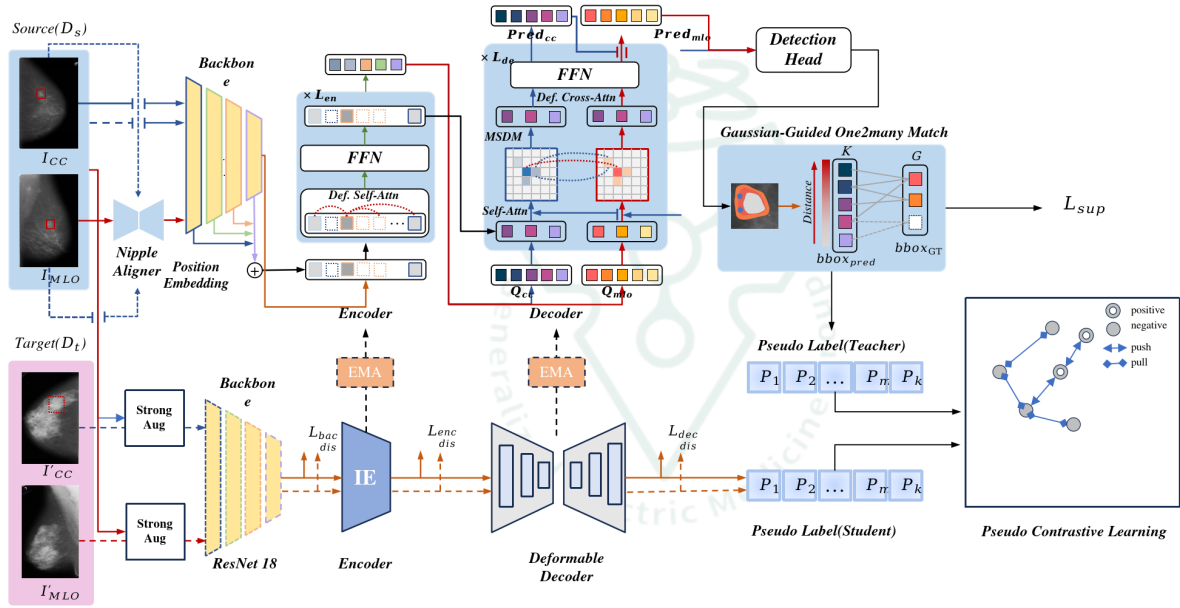
I. Summary of the progress of your research work and existing problems since last progress report

Please attach a short technical report on your research work.

The first work is to predict the mass location, so we built a multimodal model, it used text information and the image information in double view of the mammogram. After the image decoder and the text decoder, we designed a gating mechanism to contrast the positive text-image pair and the negative text-image pair.



The second work leverages mammographic images from two views and employs a graph neural network for cross-view analysis of bilateral mammograms. During the prediction-to-target matching phase, a Gaussian penalty is applied for one-to-many matching. Furthermore, to address the abundance of unlabeled data in mammography, an appropriate semi-supervised strategy has been developed.



II. Current Achievements

Please give a summary of your current achievement, e.g., research direction identified, technical reports/conference/journal papers produced or other academic activities attended.

1. Accomplish the rebuttal for the first paper, waiting for the final result.

MVCxM: Cross-view Detection in Mass Detection Mechanism based on Medical Text Message and Double View Features Revised-1st Author ACM MM

2. Accomplish the first version of the second paper, submit it in July.

III. Work Plan for Next 6 Months

Please specify in details.

No.	Content	Completion Date
1	Commercial Deployment of Mammography Workstation	2025.08
2	Waiting for the final result of the manuscript	2025.07
3	Submit the second paper about the second work	2025.07

IV. Lists

Papers Submitted/Revised/Accepted

(Past 6 months)

MVCxM: Cross-view Detection in Mass Detection Mechanism based on Medical Text Message and Double View Features Revised-1st Author

DKxMD: A Mammogram Mass Detection Framework Based on Consistent Density Map Fusion and Medical Prior Knowledge-3rd Author

(Accumulated)

Events Attended

V. Technical Report

☒ None ☐ Attached

Signature of
Student

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张宜轩 ZHANG YIXUAN

(Yixuan Zhang)

Date 2025.06.26

Opinions of Supervisor

After reviewing the semi-annual report, the Supervisor believe that,

1. The progress, the existing problems and the follow-up plan of research has been reported thoroughly.
☒ Yes ☐ No ☐ Inapplicable
2. The anticipated academic goal(s) have been achieved.
☒ Yes ☐ No ☐ Inapplicable
3. The academic outputs, e.g., publications, for the past 6 months have been as anticipated.
☒ Yes ☐ No ☐ Inapplicable
4. The work plan for next year is reasonable.
☒ Yes ☐ No ☐ Inapplicable
5. Others.
☒ None ☐ Specified below

Overall progress for the past 6 months is satisfactory.

☒ **Yes** ☐ **No**

Detail comments on overall progress:

*Signature of
Supervisor*

x

(Dr. Tao Tan)

Date